

Models as Governed Interfaces for AI-Native MBSE

Read-Side Adequacy and Write-Side Admissibility

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THE POSITION

Giving AI programmatic access to a machine-readable model is necessary, but not enough. What still needs work is the data architecture around that model.

The gap, on a real model

We use the public Apollo 11 SysML v2 reconstruction at a pinned commit (solid by current practice) and pair it with a fictional out-of-distribution control. That way we can separate training recall from invention.

Fifteen derivation-style questions; three current commercial LLMs; each run with and without a hand-written EA1–EA4 sidecar.

“Why five F-1 engines rather than four or six?”

The values themselves are in the model. What is missing is the chain that would authorise them: no traversable derivation, no epistemic status, no provenance.

AND THE NUMBERS DISAGREE

33.85 MN < 34.5 MN

Five engines at 6770 kN do not meet the stated stage minimum. The model never adds them up or raises a flag. The two sources that would close the chain, the AS-506 flight evaluation report and the 1961 Rosen committee trail, sit outside what the model can reach.

Under pressure

9/15

answers the substrate had no basis to authorise

1/45

ungrounded once we added a mocked EA1–EA4 layer

Failures under pressure roughly halved. The gain showed up as correct abstention, not as more answers. The strongest model invented most fluently, and every invented claim still sounded historically plausible.

Read side, write side

READ-SIDE ADEQUACY: what the substrate exposes
EA1 EA2 EA3 EA4

EA5 participation side

WRITE-SIDE ADMISSIBILITY: what the substrate refuses
AC1 AC2 AC3 AC4 AC5 AC6 AC7 AC8

GOVERNED SUBSTRATE: one store implements both

Epistemic adequacy

A substrate is epistemically adequate when a query can decide whether any claim it holds is grounded.

“Epistemic” here means how a claim is known, not what it says. Five criteria follow. Four show up in the case and check out independently; the fifth is a hypothesis we put forward to test.

- EA1 Derivation & rationale: the chain, and the alternatives that were weighed, are missing or stuck in free text
- EA2 Epistemic status: target, verified value, assumption and placeholder all look the same
- EA3 Provenance: who, how, review stage and evidence are not recorded
- EA4 Completeness & reachability: values needed to close a chain sit outside the model, or do not resolve
- EA5 Contribution governance [hypothesis]: nothing flags AI origin or sends the claim through review

Eight admissibility constraints

A write is blocked unless...

- AC1 the derivation points to an upstream element, or is marked unresolved
- AC2 status is taken from a closed vocabulary: no free text, no blanks
- AC3 any status downgrade records author, date and reason
- AC4 every safety-critical element has full provenance and an evidence URI that resolves
- AC5 AI-origin values enter quarantined and signed; unsigned writes are rejected
- AC6 promotion needs a review record from someone other than the proposer
- AC7 the derivation graph that results stays acyclic
- AC8 each cross-reference resolves to exactly one live element per revision

None of the mechanisms is new: PROV, NIST SP 800-53, MARTE, ISO 10007, SACM. The claim is their combination: epistemic rather than structural, attached per value, typed by AI origin, and enforced at the write gate.

What the gate cannot buy

The architecture only guarantees that every entry reaching the record has been structurally governed. It does not guarantee that the entry is correct.

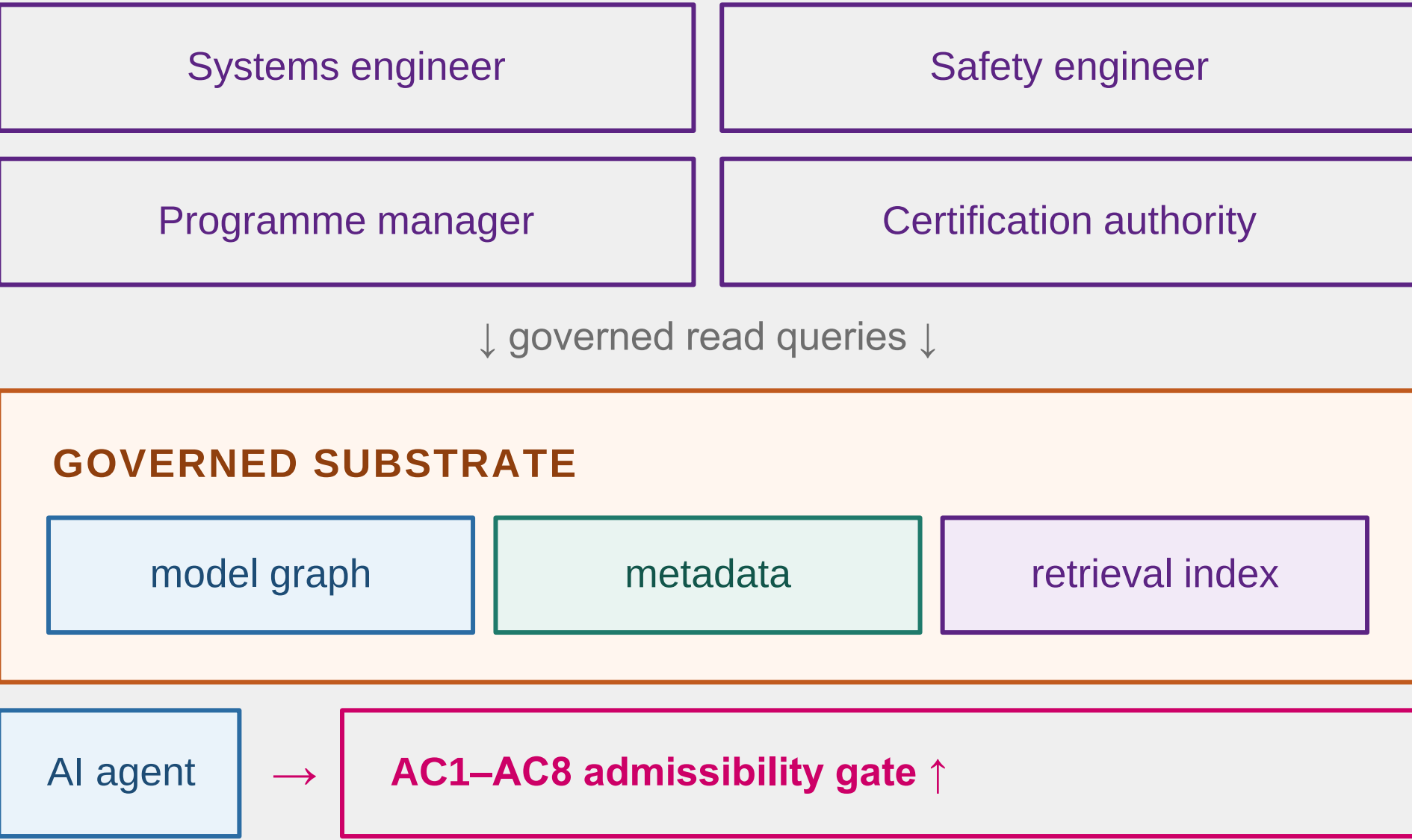
- Rubber-stamped reviews
- Coordinated falsification
- Signing-key misuse
- URIs that resolve but are wrong

A human reviewer is still the final author of any safety-relevant claim. A blocking gate also makes every write more expensive, so AC1's bar is deliberately low: mark the derivation unresolved. The evaluation then tracks how often those markers become normal practice.

Scan the QR code to follow the Research programme



Governed-Query Architecture Framework



A viewpoint is a query specification; a view is the governed result of that query. GQAF adds write-side admissibility on the same queries. Contributions only enter through the gate.

Three candidates

A: inline metadata

EA1–EA4 live on SysML v2 elements; the gate sits at the API. Durable for archive, but standards move slowly, and native writes can bypass the gate.

B: metadata graph

A separate graph keyed by programme-stable identifiers, with the gate on the graph side. Deployable today; EA4 gaps become queries. Two systems of record can drift; content hashes make that visible. Our recommended first step.

C: native store

One transactional engine holds topology, metadata and index. Higher cost and a larger failure surface. This is the long-term build target, not a product yet.

A falsifiable plan

Model and prompt stay fixed across three arms: retrieval over unstructured content; the structured model with epistemic metadata stripped out; and the structured model with that metadata present. The middle arm is the null: structure alone already improves precision.

Metrics: groundedness, correct abstention, completeness of the derivation chain, and fidelity of contribution governance.

If, across two cycles, the substrate cannot beat the stripped arm on correct abstention and governance fidelity, we treat the claim as refuted.

Try to break it

Which criterion would you drop, and what ungrounded answer then survives?

Where would a blocking write gate cost too much in your practice?

Bring a model and a derivation question we did not pick.